Atomic structure

Read the Resource sheet 'An introduction to atomic structure', then answer the following questions.

☞ **1** Use the words in the word bank to complete the following passage. Some words will be used more than once.

Word bank
three
atom
electrons
nucleus
substances
100
neutrons
spinning
one
copper
protons
unique

Elements are simple s_____ made up of the same type of a_____. There are more than 1_____ different types of elements. Each element's atom is u_____. For example, c_____ atoms are different from silver atoms.

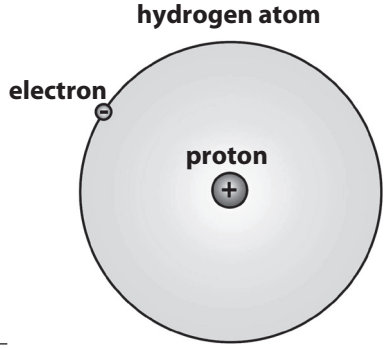
Atoms are made up of t_____ different sub-atomic particles: n_____, protons and e_____. The middle part of the atom is called the n_____. The nucleus contains p_____ and neutrons. The electrons are found s_____ around the outside of the n_____.

An atom has the same number of protons and e_____. For example, hydrogen has one proton, so it also has o_____ electron.

2 Match the sentence halves to complete the information below.

Protons and neutrons are	no charge. This means they are neutral.
Protons have a	can be found spinning around the nucleus.
Electrons are tiny and	positive charge.
Neutrons have	about the same size and mass.
Electrons have	a negative charge.

3 Look at the diagram of a hydrogen atom.



a How many protons does a hydrogen atom have? _____

b How many neutrons does a hydrogen atom have? _____

c How many electrons does a hydrogen atom have? _____

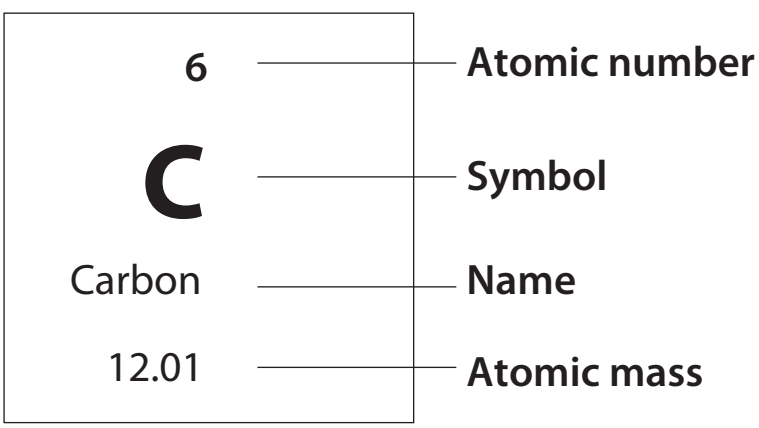
d What is unusual about hydrogen? _____

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ISBN 978 1 92571 062 5

Atomic structure and the periodic table (1)

The **periodic table** is a record of all the **elements**. The periodic table gives us lots of information about each element. It tells us:

- each element's symbol
- the number of **protons** in an element's atom. This is called its **atomic number**. The atomic numbers increase by one as you go along the periodic table.
- the **atomic mass** of an element. This tells us the mass of an element compared to carbon. It is approximately the total number of protons and **neutrons** added together.
- the number of **electrons** in an element's atom. The atomic number tells us this, too, because this number is always the same as the number of protons.
- some of the properties of different elements.



From looking at the information provided for carbon, we know that carbon:

- has the symbol C
- has an atomic number of 6. This means it has 6 protons and 6 electrons.
- has an atomic mass of 12. This means it has 6 neutrons (atomic mass minus the number of protons).

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Atomic structure and the periodic table (2)

Read the Resource sheet 'Atomic structure and the periodic table (1)' and then answer the following questions.

1	6	8
H	C	O
Hydrogen	Carbon	Oxygen
1.01	12.01	16.00

☞ **1** The diagram above is from the periodic table. Use the information provided to complete the table below. Round to the nearest whole number.

	Hydrogen	**Carbon**	**Oxygen**
Symbol			
Atomic number			
Atomic mass			
Number of protons			
Number of electrons			
Number of neutrons			

2 An atom contains 28 protons and 31 neutrons.

a What is its atomic number? _____

b How many electrons does it have? _____

c What is the atomic mass of the atom? _____

d Use the periodic table on the inside front cover to find the name of the element. _____

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